

8-BIT DIVISION

EXP NO: 4

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

A03

BC0203

DE0000

HL0000

PSW0000

PC421B

SPFFFF

Int-Reg00

Flag

S1

Z0

AC0

P0

C1

Load me at

1START: NOP

2LDA 8500

3MOV B, A

4LDA 8501

5MVI C, 00H

6LOOP: CMP B

7JC LOOP1

8SUB B

9INR C

10JMP LOOP

11LOOP1: STA 8502

12MOV A, C

13STA 8503

14RST 1

15HLT

Decimal - Hex Conversion

Decimal

Hex

0

0

To Hex

To Dec

I/O Ports

0

-

+

00

Update Port Value

Memory

8501

-

+

06

Update Memory

Data

Stack

KeyPad

Memory

I/O Ports

Start8500

OK

Address (Hex)

Address

Data

213485002

213585016

213685020

213785033

213885040

213985050

213A85060

213B85070

213C85080

213D85090

213E85100

213F85110

214085120

214185130

Line No

Assembler Message

0Program assembled successfully

Simulator: Idle

16-BIT SUBTRACTION

EXP NO: 6

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

A	01
BC	03 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 13
SP	FF FF
Int-Reg	00

Flag

S	0
Z	0
AC	0
P	0
C	1

Load me at

1 LDA 2050

2 MOV B, A

3 LDA 2052

4 SUB B

5 CMA

6 INR A

7 STA 2054

8 MVI A, 01

9 STA 2055

10 HLT

Data Stack Keypad Memory I/O Ports

Start 2050 OK

Address (Hex)	Address	Data
0802	2050	3
0803	2051	0
0804	2052	2
0805	2053	0
0806	2054	1
0807	2055	1
0808	2056	0
0809	2057	0
080A	2058	0
080B	2059	0
080C	2060	0
080D	2061	0
080E	2062	0
080F	2063	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

16-BIT MULTIPLICATION

EXP NO: 7

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers Flag Load me at

Register	Value	Flag
A	00	S 0
BC	00 00	Z 1
DE	00 00	AC 0
HL	00 00	P 1
PSW	00 00	C 0
PC	42 22	
SP	19 0A	
Int-Reg	00	

Decimal - Hex Conversion

Decimal	Hex
0	0

To Hex To Dec

I/O Ports

Port	Value
0	00

Update Port Value

Memory

Address	Value
2052	05

Update Memory

1 LHL 2050
2 SPHL
3 LHL 2052
4 XCHG
5 LXI H,0000H
6 LXI B,0000H
7 AGAIN: DAD SP
8 JNC START
9 INX B
10 START: DCX D
11 MOV A,E
12 ORA D
13 JNZ AGAIN
14 SHLD 2054
15 MOV I,C
16 MOV H,B
17 SHLD 2055
18 HLT

Data Stack KeyPad Memory I/O Ports

Start 2050 OK

Address (Hex)	Address	Data
0802	2050	10
0803	2051	25
0804	2052	5
0805	2053	0
0806	2054	50
0807	2055	0
0808	2056	0
0809	2057	0
080A	2058	0
080B	2059	0
080C	2060	0
080D	2061	0
080E	2062	0
080F	2063	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

SWAPING OF TWO 8-BIT DATA

EXP NO: 10

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers Flag Load me at

Register	Value	Flag
A	06	S 0
BC	06 04	Z 1
DE	00 00	AC 0
HL	00 00	P 1
PSW	00 00	C 0
PC	42 10	
SP	FF FF	
Int-Reg	00	

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

1101 - + 04

Update Memory

1 LDA 1100
2 MOV B,A
3 LDA 1101
4 MOV C,A
5 STA 1102
6 MOV A,B
7 STA 2203
8 HLT

Start 1100 OK

Address (Hex)	Address	Data
044C	1100	6
044D	1101	4
044E	1102	4
044F	1103	6
0450	1104	0
0451	1105	0
0452	1106	0
0453	1107	0
0454	1108	0
0455	1109	0
0456	1110	0
0457	1111	0
0458	1112	0
0459	1113	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle